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Analysis of idealized Zeeman quantum beat experiments in the advanced laboratory

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An expression for the time-dependent intensity of light scattered in idealized Zeeman quantum beat experiments commonly found in advanced teaching laboratories is presented. The development of this expression illustrates the separation of geometrical and dynamical factors via the Wigner-Eckart theorem. The experiment is also interpreted in terms of a pictorial description of a rotating electron cloud, and an experimental procedure to illustrate this rotation is described. © 1996 American Association of Physics Teachers.

I. INTRODUCTION

The interference of probability amplitudes is one of the most fundamental and distinguishing features of quantum mechanics.¹ An interesting example of this effect can occur when atomic states are excited simultaneously. If these states have slightly different energies and yet decay to the same final state, their time evolution at different rates produces interference effects that cause the intensity and polarization of the subsequent decay radiation to exhibit an oscillatory time dependence commonly known as “quantum beats.” Often, an externally applied magnetic field is used to produce the energy splitting via the Zeeman effect. Numerous investigations of quantum beats have been reported.^{2–4}

Brandenberger^{5,6} has described experimental investigations of Zeeman quantum beats in neon that are well suited for use in the advanced teaching laboratory. In this report, we present an analysis of these experiments by developing an expression that may be used to model the time-dependent intensity of the scattered radiation under idealized conditions. In particular, this development illustrates the separation of geometrical and dynamical factors via the Wigner-Eckart theorem and is suitable for use in the graduate laboratory. We also develop a “pictorial” model in which the electron cloud of the excited system rotates in space. This model is useful for describing the quantum beat mechanism in the undergraduate laboratory without the need to rely on more advanced angular momentum theory.

The experimental setup is shown in Fig. 1. Briefly, the beam from a nitrogen-pumped, tunable dye laser is directed into a cell containing neon gas at a pressure of 500 mTorr or less. A weak radio frequency discharge in the cell maintains a population of atoms in the metastable $3s'_{00}$ (systematic notation⁷) or $2s_3$ (Paschen notation⁷) state of the $1s^2 2s^2 2p^5 3s$ manifold. The laser is tuned to a wavelength of 626.6 nm to excite some of these atoms to the $3p'_{11}$ or $2p_5$ state of the $1s^2 2s^2 2p^5 3p$ manifold. Most of these atoms decay back to the $3s'_{00}$ state. The 626.6 nm decay radiation passes through a linear polarizer, interference filter, and fiber optic before being detected by a photomultiplier tube wired for fast response.

The laser pulse width is on the order of 5 ns. Because this time is reasonably short compared to the 18.8 ns lifetime⁶ of the $3p'_{11}$ state, it is possible to observe the exponential decay of the excited atoms. When a magnetic field is applied, the $3p'_{11}$ state undergoes Zeeman splitting so that the excited atoms are formed in a superposition of magnetic sub-

states with different energies. These states evolve at different rates, and the emitted radiation exhibits time-dependent oscillations or quantum beats. A digital oscilloscope is used to average the signal over several laser pulses. A typical spectrum (locus of open circles) is shown in Fig. 2.

II. ANALYSIS

We use the coordinate system shown in Fig. 1 to analyze the experiment. The laser beam is directed along the y axis and is linearly polarized along the x axis. Radiation emitted in the z direction passes through a linear polarizer whose transmission axis is oriented at an angle α relative to the x axis before being detected. The magnetic field applied to the cell is oriented in the z direction.

The spectral width of the laser pulse is on the order of 2 GHz. For the 10–100 Gauss magnetic fields used in the experiment, the Zeeman splitting is less than 0.3 GHz, so the splitting is sufficiently small that it may be neglected in the excitation process. The amplitude for excitation from the initial state $|jm\rangle = |00\rangle$ into each final substate $|jm\rangle = |1m\rangle$ is proportional to the dipole matrix element $\langle 1m|\epsilon_l \cdot \mathbf{r}|00\rangle$ where ϵ_l is the polarization vector of the laser beam. Just after the absorption of the photon, the excited state $|t=0\rangle$ is described by the superposition

$$|t=0\rangle = A \sum_m |1m\rangle \langle 1m|\epsilon_l \cdot \mathbf{r}|00\rangle, \quad (1)$$

where A is a proportionality constant.

The key factor in the analysis presented here is that it is not necessary to know the complete wave function which describes the excited state to analyze the experimental measurements. The Wigner-Eckart theorem⁸ allows the dynamical factors associated with the photon interaction between the initial and final states to be separated from the geometrical factors associated with the different magnetic substates. In other words, the relative amplitudes and phases of the excited substates in Eq. (1) may be determined simply from the angular momentum properties of these substates.

For our case, $\epsilon_l = \hat{x}$ so that

$$\epsilon_l \cdot \mathbf{r} = x = -\frac{1}{\sqrt{2}}(r_{+1} - r_{-1}), \quad (2)$$

where $r_{\pm 1}$ are spherical components⁸ of $\mathbf{r}$,

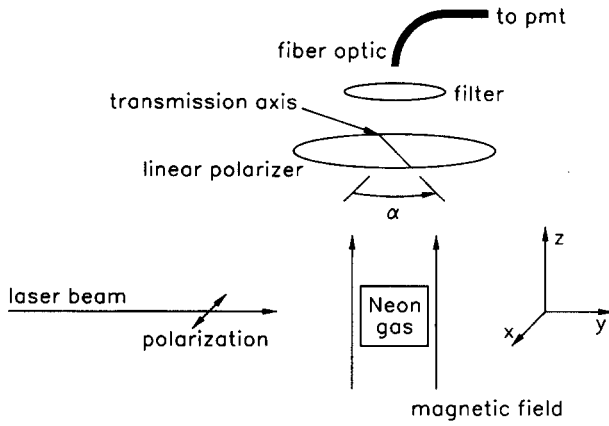


Fig. 1. Schematic diagram of the experiment and coordinate system used in the analysis.

$$r_{\pm 1} = \mp \frac{(x \pm iy)}{\sqrt{2}} \quad \text{and} \quad r_0 = z.$$

Using the Wigner–Eckart theorem, the matrix elements of the spherical components of $\mathbf{r}$ between the initial and excited states may be expressed as

$$\langle 1m | r_q | 00 \rangle = (-1)^{1-m} \begin{pmatrix} 1 & 1 & 0 \\ -m & q & 0 \end{pmatrix} \langle 1 || \mathbf{r} || 0 \rangle, \quad (3)$$

where $\langle 1 || \mathbf{r} || 0 \rangle$ is the reduced matrix element between the initial and excited states and $\begin{pmatrix} 1 & 1 & 0 \\ -m & q & 0 \end{pmatrix}$ is a Wigner $3j$ symbol.⁹ The reduced matrix element contains the dynamical factors which depend on the explicit form of the initial and final state wave functions. The relative amplitudes and phases of the excited substates depend on the angular momentum properties of the excited substates and are determined by the sign and $3j$ symbol.

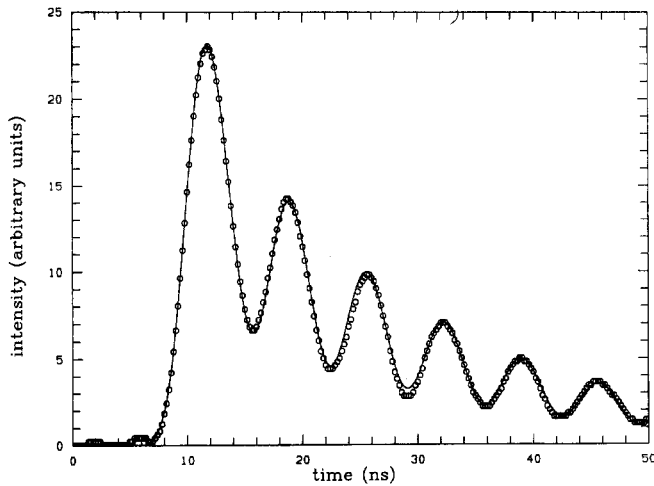


Fig. 2. Intensity vs time spectrum collected with a polarizer position $\alpha = 45^\circ = 0.785$ rad. The line is a fit to the measurements using Eqs. (15) and (12). As described in Sec. III, the variable α is treated as a parameter in the fit and interpreted as the rotation angle of the electron cloud ωt_e . The fit yields $I_{\text{reflected}} = 7.3 \pm 0.7$, $C = (8.8 \pm 0.3) \times 10^9$, $\Gamma = (5.15 \pm 0.11) \times 10^7 \text{ s}^{-1}$, $a = (1.93 \pm 0.04) \text{ ns}$, $t_0 = (10.5 \pm 0.1) \text{ ns}$, $\omega = (4.69 \pm 0.02) \times 10^8 \text{ s}^{-1}$, and $\omega t_e = 0.79 \pm 0.06$.

Evaluating the $3j$ symbol for each excited substate gives contributions only when $m = \pm 1$, and the excited state in Eq. (1) can be written as

$$|t=0\rangle = -\frac{A}{\sqrt{6}} \langle 1 || \mathbf{r} || 0 \rangle (|11\rangle - |1-1\rangle). \quad (4)$$

The time evolution of each excited substate is given by the standard exponential factors $e^{-iEt/\hbar}$ where E is the energy of the substate. This energy may be expressed as the sum of the unperturbed energy E_0 and two other terms. First, the Zeeman splitting of the excited state is produced by the interaction of the magnetic moment μ of the excited state and the applied magnetic field $\mathbf{B} = B\hat{z}$. The magnetic moment may be expressed in terms of the Bohr magneton μ_B , the angular momentum $\mathbf{J}$, and the Landé g factor of the excited state,

$$\Delta E = -\mu \cdot \mathbf{B} = -\left(\frac{-g\mu_B \mathbf{J}}{\hbar}\right) \cdot \mathbf{B} = g\mu_B m B = m\hbar\omega, \quad (5)$$

where $\omega = g\mu_B B/\hbar$ is the angular frequency corresponding to the energy difference between adjacent magnetic sublevels. Second, the term $-i\hbar\Gamma/2$ is added to the energy to account for the decay of the excited state with decay constant Γ . Combining the time evolution factors with the initial state in Eq. (4) gives the time evolved state $|t\rangle$,

$$|t\rangle = -\frac{A}{\sqrt{6}} \langle 1 || \mathbf{r} || 0 \rangle [|11\rangle e^{-i(E_0 + \hbar\omega)t/\hbar} - |1-1\rangle e^{-i(E_0 - \hbar\omega)t/\hbar}] e^{-\Gamma t/2}. \quad (6)$$

The decay radiation emitted with polarization ϵ is found from the dipole matrix element $\langle 00 | \epsilon \cdot \mathbf{r} | t \rangle$ between the time evolved excited state and the final $3s'_0$ state. For the polarizer oriented at an angle α relative to the x axis,

$$\epsilon \cdot \mathbf{r} = \frac{1}{\sqrt{2}} (r_{-1} e^{i\alpha} - r_{+1} e^{-i\alpha}), \quad (7)$$

and we may evaluate the matrix element by using the Wigner–Eckart theorem to separate the geometrical and dynamical factors,

$$\langle 00 | \epsilon \cdot \mathbf{r} | t \rangle = \frac{A}{3} |\langle 0 || \mathbf{r} || 1 \rangle|^2 e^{-iE_0 t/\hbar} e^{-\Gamma t/2} \cos(\omega t - \alpha). \quad (8)$$

The intensity of the emitted radiation is proportional to the squared magnitude of the dipole matrix element,

$$I(t) \propto |\langle 00 | \epsilon \cdot \mathbf{r} | t \rangle|^2 \propto e^{-\Gamma t} \cos^2(\omega t - \alpha). \quad (9)$$

Equation (9) applies to the case of an infinitesimally short laser pulse. It cannot be used to analyze the experimental measurements in actual cases since the source consists of atoms excited at different times t_e throughout the duration of the laser pulse. We assume that the number of atoms dN excited in the time interval between t_e and $t_e + dt_e$ is proportional to a Gaussian distribution centered at time t_0 ,

$$dN \propto e^{-(t_e - t_0)^2/a^2} dt_e, \quad (10)$$

where a is a measure of the temporal width of the source. Then the time in Eq. (9) is replaced by the time $t - t_e$ after excitation and the total intensity is found by integrating over all excitation times prior to the observation time,

$$I(t) = C \int_{-\infty}^t e^{-(t_e - t_0)^2/a^2} e^{-\Gamma(t-t_e)} \cos^2[\omega(t-t_e) - \alpha] dt_e, \quad (11)$$

where C is a proportionality constant. This integration gives

$$I(t) = C \left(\frac{\sqrt{\pi}a}{2} e^{a^2\Gamma^2/4} e^{-\Gamma(t-t_0)} \times \{1 + e^{-\omega^2 a^2} \cos[2\omega(t-t_0) - 2\alpha - \Gamma\omega a^2]\} - \frac{\sqrt{\pi}a}{4} e^{-(t-t_0)^2/a^2} \times \{W(iu) + \text{Re}[e^{2i\alpha}W(\omega a + iu)]\} \right), \quad (12)$$

where

$$u = \frac{t-t_0}{a} - \frac{\Gamma a}{2} \quad (13)$$

and $W(z)$ is a standard function¹⁰ related to the error function of complex argument,

$$W(z) = e^{-z^2} [1 - \text{erf}(-iz)]. \quad (14)$$

Equation (12) divides into two parts. For $u > 0$, that is, for $t > t_0 + \Gamma a^2/2$, the magnitude of the W function is less than unity and the factor of $e^{-(t-t_0)^2/a^2}$ allows the second part of Eq. (12) to be neglected soon after the laser pulse. In this case, the intensity is given by the first part only and has the expected form of a constant plus oscillatory term multiplied by the exponential decay. Note that the finite width a of the laser pulse reduces the amplitude or “depth” of the oscillatory term and also shifts the phase of the oscillations. Note also that in the ideal case of $a \approx 0$, $\exp(-\omega^2 a^2) \approx 1$ and the “modulation depth” of the beats is 100%. Before the laser pulse, the contributions from the first and second parts in Eq. (12) are equal in magnitude and cancel.

Nonlinear least-squares fitting may be used to fit Eq. (12) to the experimental measurements. It is possible to fit only the measurements after the laser pulse using just the first part of Eq. (12) or all of the measurements, including those during the laser pulse, using the entire expression. If measurements during the laser pulse are included, it is desirable to add a term proportional to the laser pulse to account for laser light reflected directly into the detector

$$I_{\text{fit}}(t) = I_{\text{reflected}} e^{-(t-t_0)^2/a^2} + I_{\text{signal}}(t), \quad (15)$$

where $I_{\text{reflected}}$ is a constant and $I_{\text{signal}}(t)$ is the intensity given in Eq. (12).

Typically the parameters $I_{\text{reflected}}$, C , a , Γ , t_0 , and ω are adjusted in the fitting procedure. As discussed in Sec. III, it may also be desirable to treat the polarizer position α as a parameter and to determine its value in the fit. Results of this fitting procedure are shown as the solid line in Fig. 2. In particular, the decay constant $\Gamma = (5.15 \pm 0.11) \times 10^7 \text{ s}^{-1}$ gives a lifetime $\tau = 1/\Gamma$ of $(19.4 \pm 0.4) \text{ ns}$ which is in rough agreement with the previously measured⁶ value of $(18.8 \pm 0.2) \text{ ns}$. Using the relation $\omega = g\mu_B B/\hbar$, these results also give $g = 1.01 \pm 0.04$ where the uncertainty in g is determined primarily by the uncertainty of the magnetic field strength. This result is in agreement with the previously measured⁵ value of 0.994 ± 0.002 .

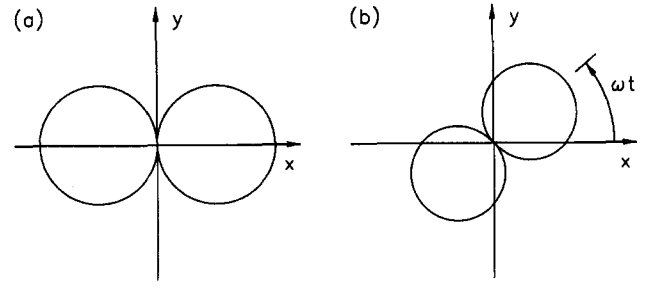


Fig. 3. Probability distribution in the x - y plane of the electron cloud describing the excited state for (a) $t=0$ and (b) $t>0$. The distribution peaks at $\phi = \omega t$ and rotates about the z axis with angular frequency $\omega = g\mu_B B/\hbar$.

It is important to note that at typical cell pressures of 100–500 mTorr, atomic collisions in the cell will shorten the lifetime of the excited atoms. These collisions can also induce transitions between magnetic sublevels, thereby affecting the polarization of emitted radiation. These effects can be taken into account⁶ by performing measurements as a function of the cell pressure and extrapolating these measurements to zero pressure. Because the analysis presented here has neglected collisions, and because of the assumption of a Gaussian profile in Eq. (10), the lineshape given in Eq. (12) can only be interpreted as an idealized lineshape.

III. PICTORIAL MODEL

As discussed in Sec. II, the excited atoms are described by a superposition of the $|1+1\rangle$ and $|1-1\rangle$ states. Because of the Zeeman splitting, these states evolve at different rates causing the emitted radiation to exhibit quantum beats.

It is helpful to interpret these results in terms of a model based on the electron probability distribution in the atom. Even though the orbital angular momentum of the excited electron is not described by a good quantum number, it is convenient to construct an approximate wave function to represent the electron, and we describe the excited $3p$ electron using spherical harmonics with $l=1$. Then the wave function corresponding to the initial state in Eq. (4) is given by

$$\psi = R(Y_{1+1} - Y_{1-1}) = -R \sqrt{\frac{3}{2\pi}} \sin\theta \cos\phi, \quad (16)$$

where R includes the radial part of the wave function and constant factors. The probability distribution is given by $\psi^* \psi$,

$$\psi^* \psi = R^2 \frac{3}{2\pi} \sin^2\theta \cos^2\phi. \quad (17)$$

Figure 3(a) shows the angular dependence of this probability distribution in the x - y plane. The distribution peaks at $\phi=0$ and the electron cloud is aligned with the laser polarization along the x axis. In fact, this alignment may be used to justify the relative minus sign between the two terms in Eq. (16) instead of relying on the angular momentum theory presented in Sec. II.

The time evolved wave function includes the time evolution factors for each substate as in Eq. (6),

$$\psi(t) = R(Y_{1+1} e^{-i(E_0 + \hbar\omega)t/\hbar} - Y_{1-1} e^{-i(E_0 - \hbar\omega)t/\hbar}) e^{-\Gamma t/2}, \quad (18)$$

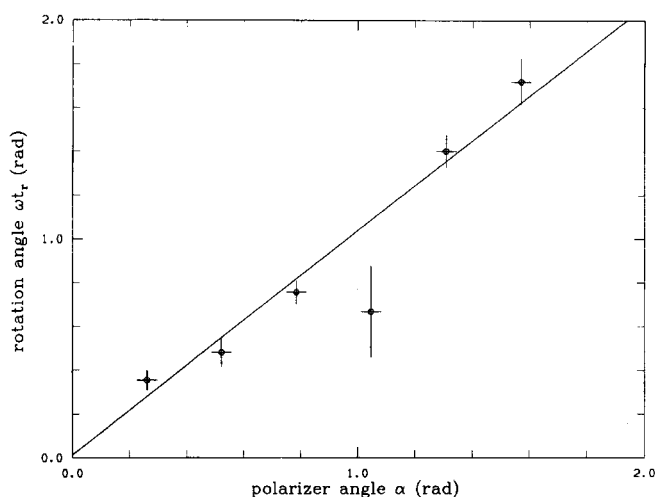


Fig. 4. Measured rotation angle ωt_r vs polarizer angle α . The fit is a straight line with slope 1.03 ± 0.07 and intercept 0.01 ± 0.06 .

and the probability distribution is given by

$$\psi^* \psi = R^2 \frac{3}{2\pi} e^{-\Gamma t} \sin^2 \theta \cos^2(\phi - \omega t). \quad (19)$$

The angular dependence of this probability distribution in the x - y plane is plotted in Fig. 3(b). The distribution peaks at an angle $\phi = \omega t$, so the electron cloud rotates about the z axis with angular velocity $\omega = g\mu_B B/\hbar$.

The intensity and polarization of the emitted radiation are determined by the size and orientation of the electron cloud. The intensity of the total radiation emitted from the source is proportional to the integrated probability distribution and decreases exponentially in time. The emitted radiation is linearly polarized along the symmetry axis of the electron cloud, that is, along the axis $\phi = \omega t$. The fraction of radiation transmitted by the linear polarizer depends on the angle $\omega t - \alpha$ between the polarization direction of the radiation and the transmission axis of the polarizer,

$$I_{\text{transmitted}} = I_{\text{incident}} \cos^2(\omega t - \alpha) \propto e^{-\Gamma t} \cos^2(\omega t - \alpha) \quad (20)$$

in agreement with Eq. (9). Integration over the laser pulse gives Eq. (12).

We can “observe” the rotation of the electron cloud experimentally by changing the position of the polarizer and comparing its position with the phase of the quantum beats. Since the transmitted intensity is greatest when the electron cloud is aligned with the transmission axis of the polarizer, rotating the polarizer to positions with $\alpha > 0$ will cause the peaks of the quantum beats to occur at later times. This effect may be seen directly in a series of spectra similar to Fig. 2 collected with different polarizer angles.

It is convenient to analyze such spectra by fitting a slightly modified form of Eq. (12) to the experimental measurements. We remove the term involving α and add a term to account

for a rotation time t_r between the excitation of the source and the alignment of the source with the polarizer. Then the argument of the cosine function is written as $[2\omega(t - t_0 - t_r) - \Gamma\omega a^2]$ and the product ωt_r represents the angle the electron cloud rotates before it is aligned with the polarizer. As mentioned at the end of Sec. II, this procedure is equivalent to treating the variable α as a parameter in the fit and interpreting its value as the rotation angle ωt_r . Figure 4 shows measurements which illustrate the expected linear relationship between the rotation angle and the polarizer angle.

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¹For example, R. P. Feynman, R. B. Leighton, and M. Sands, *The Feynman Lectures on Physics*, Vol. III (Addison Wesley, Reading, MA, 1965), Chap. 1.

²S. Haroche, “Quantum Beats and Time-Resolved Fluorescence Spectroscopy,” in *High-Resolution Laser Spectroscopy*, edited by K. Shimoda (Springer-Verlag, Berlin, 1976), pp. 253–313.

³J. N. Dodd and G. W. Series, “Time-Resolved Fluorescence Spectroscopy,” in *Progress in Atomic Spectroscopy Part A*, edited by W. Hanle and H. Kleinpoppen (Plenum, New York, 1978), pp. 639–677.

⁴R. Gupta, “Resource Letter LS-1: Laser Spectroscopy,” *Am. J. Phys.* **59**, 874–886 (1991), Sec. V D.

⁵J. R. Brandenberger, “Measurement of g -Factors in Neon by Quantum Beat Spectroscopy,” *Opt. Comm.* **30**, 181–183 (1979); see also J. R. Brandenberger, *Lasers and Modern Optics in Undergraduate Physics, A Report to Foundations, Corporations, and Undergraduate Colleges* (Lawrence University, Appleton, WI, 1989), pp. 59–65.

⁶J. R. Brandenberger and B. R. Rose, “Quantum Beat Determination of Lifetimes and Disalignment Cross Sections in Neon,” *Opt. Comm.* **36**, 453–459 (1981).

⁷H. M. Crosswhite and G. H. Dieke, “Important Atomic Spectra,” in *American Institute of Physics Handbook*, edited by D. E. Gray (McGraw-Hill, New York, 1972), 3rd ed., Sec. 7e. Paschen notation is a semiempirical notation that is used extensively for rare-gas spectra and labels states within a manifold by enumeration. In systematic notation for rare-gas atoms, the valence electron is specified by its nl symbol. The first subscript is the angular momentum of the atom exclusive of the spin of the electron, minus 1/2. The second subscript is the total angular momentum J of the atom. The levels are primed if they converge to the $^2P_{1/2}$ level of the core. Then the lower $3s'_{00}$ state is described approximately as a $3s$ valence electron and $1s^2 2s^2 2p^5(^2P_{1/2})$ core with $j-l$ coupling between core and electron equal to 1/2 and total angular momentum $J=0$. Similarly, the excited $3p'_{11}$ state is described approximately as a $3p$ valence electron and $1s^2 2s^2 2p^5(^2P_{1/2})$ core with $j-l$ coupling between core and electron equal to 3/2 and total angular momentum $J=1$.

⁸For example, A. R. Edmonds, *Angular Momentum in Quantum Mechanics* (Princeton University, Princeton, NJ, 1974), 2nd ed., 3rd printing with corrections, Chap. 5.

⁹For example, Chap. 3 of Ref. 8.

¹⁰W. Gautschi, “Error Function and Fresnel Integrals,” in *Handbook of Mathematical Functions*, edited by M. Abramowitz and I. A. Stegun (Dover, New York, 1972), Chap. 7.